

Aleksa Vukadinović

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EDUCATION

Faculty of Mathematics, University of Belgrade

Oct 2022 – Present

B.S. Computer Science

Faculty of Mathematics, Belgrade

- **Courses:** Algorithms and Data Structures, Construction and Analysis of Algorithms, Computer Graphics, Introduction to Web and Internet technologies, Interpretator design, and many more...
- **Interests:** Algorithms and Data Structures, Computer Graphics, AI, Compilers & Interpreters

TECHNICAL SKILLS

Programming Languages: C/C++, JavaScript, TypeScript, Python, Java, C#, Haskell

Web Technologies: React, Next.js, Node.js, Fastify, HTML, CSS

Tools & Technologies: OpenGL, PostgreSQL, MongoDB, Docker, Git, Linux, LaTeX, Postman

Languages: Serbian, English, Italian

EXPERIENCE

Junior Data Engineer

Nov 2025 – Present

Nordeus

Belgrade, Serbia

- Designed and maintained data pipelines using **Apache Airflow** for orchestration and scheduling
- Worked with event-driven architectures using **Apache Kafka** and **Google Pub/Sub**
- Built and optimized **ETL** processes for large-scale analytics workloads
- Queried and analyzed data in **Google BigQuery** to support business and game analytics
- Collaborated with analytics and backend teams to ensure data reliability and scalability

Full-Stack Developer Intern

Feb 2025 – Aug 2025

Novet.ai

Belgrade, Serbia

- Developed internal dashboards and production features using **TypeScript**, **React**, and **Next.js**
- Built and optimized REST APIs with **Fastify** (TypeScript) and **Flask** (Python)
- Worked with SQL databases, performed migrations, and improved backend performance
- Implemented an AI-powered document redlining system for detecting and grouping text changes
- Collaborated closely with product and design teams on internal tools and workflows

PROJECTS

Starfall Shores | *Computer Graphics course project*

Jun 2025

- Created a visually rich real-time rendering scene using **C++**, **OpenGL** and a **custom-built engine**
- Wrote custom **GLSL shaders** for lighting, materials, and post-processing effects
- Implemented advanced graphics techniques such as **instanced rendering** for performance optimization and **bloom** for enhanced visual fidelity
- Implemented a dynamic day/night cycle with smooth real-time transitions in **C++/OpenGL**

DigitEye | *AI neural network for image recognition*

Nov 2024

- Python neural network based on **TensorFlow** and **Keras** trained on the MNIST dataset for recognizing handwritten digits.
- Neural network can successfully identify the digit in 90% of the cases, also supports custom parameters, such as number of layers, epochs, etc.

Volley of Rockets in a Tornado | *Python simulation*

Jun 2024

- Collaborated with a team of 2 and developed a fully functional simulation in **Python** simulating trajectories of projectiles launched at a tornado with varying angles and velocities.
- Project was done as a part of Introduction to Mechanics course in second year

AmbroAI | *AI model for predicting number of allergic reactions in population*

Sep 2024

- Developed an AI model based on linear regression in **Python** to predict the number of people likely to have an allergic reaction, based on geographic location and population size.
- This project was developed during 'The Great Hacka' hackathon.